

Dynamic Log Monitoring in Kubernetes Applications with Lightrun

Embrace the potential of Lightrun, a powerful logging and monitoring framework, to streamline operations, reduce maintenance efforts, and gain deeper insights into the health and performance of your Kubernetes applications in the cloud.



Image Source: <https://lightrun.com>

in visually interpreting monitoring logs through a dashboard to identify anomalous behaviour or check the health of an application.

This is more complex in container applications in cloud platforms (private cloud or public cloud). Real-time monitoring of health-checks, status, logs and even notification of Kubernetes services, along with getting diagnostic information in real-time, are common features of cloud services. Designing an observability pattern led solution for the monitoring and management of services requires quick analysis of critical issues on any cloud platform. Infrastructure monitoring and observability are important for any cloud architecture; these must provide a solution that caters to business goals while ensuring cost-effective maintenance and operational support.

Lightrun is a powerful logging and monitoring framework, which can be used to dynamically add debug/log statements in running Java applications. It also allows to add metrics, logs and traces to monitor trends in the application's running state of defined expressions. In a production running application, this kind of dynamic monitoring service helps to troubleshoot applications in their 'as is' condition without altering the code by adding debug statements and rebuilding/redeploying applications. Another key

Observability is very important to understand the behaviour of cloud resources and monitor their health. There are two options available for choosing observability services – native services provided by cloud service providers (like Azure, AWS and GCP) and third party services like Lightrun, Grafana and Prometheus. Third party tools are cloud-agnostic and can be easily integrated into the cloud services.

Real-time monitoring and tracking of resources plays an important role

in cloud architecture as it helps to understand the behaviour, health status and utilisation of cloud services easily as well as to drive CloudOps. Operational metrics in a cloud environment, like logs and usage traces, are often high in volume making their analysis a tedious task. However, analysing, managing and monitoring these metrics can provide valuable insights into performance, usage patterns and application/infrastructure health. Observability tools like Lightrun prove to be handy